

TIC 313485023

PERIOD ALIAS

Eclipsing binary candidate dossier - Star Vetter

BLS EPHEMERIS

Period 6.576156 d (ALIASED, see below)
-> corrected period: 13.152313 d
Epoch (t0) 1712.1885 BTJD
Duration 0.2440 d
Depth 1.283 % (12,831 ppm)
SNR 212.8

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

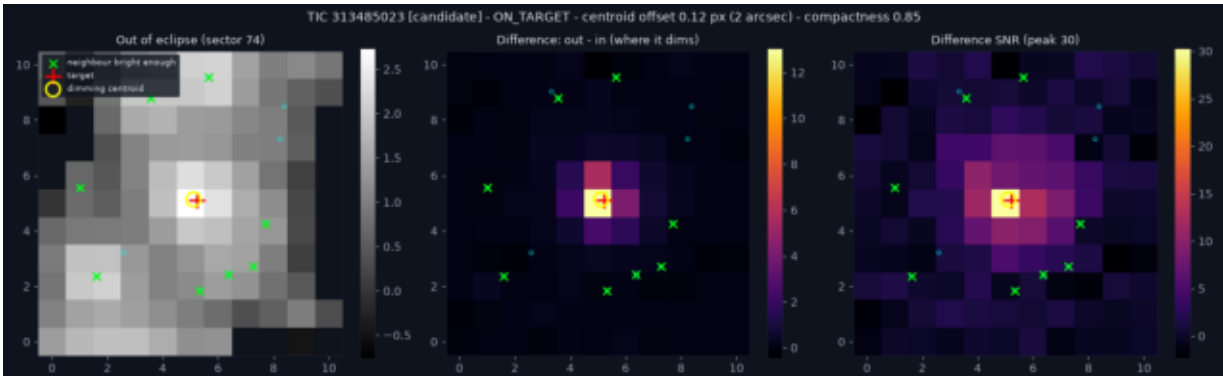
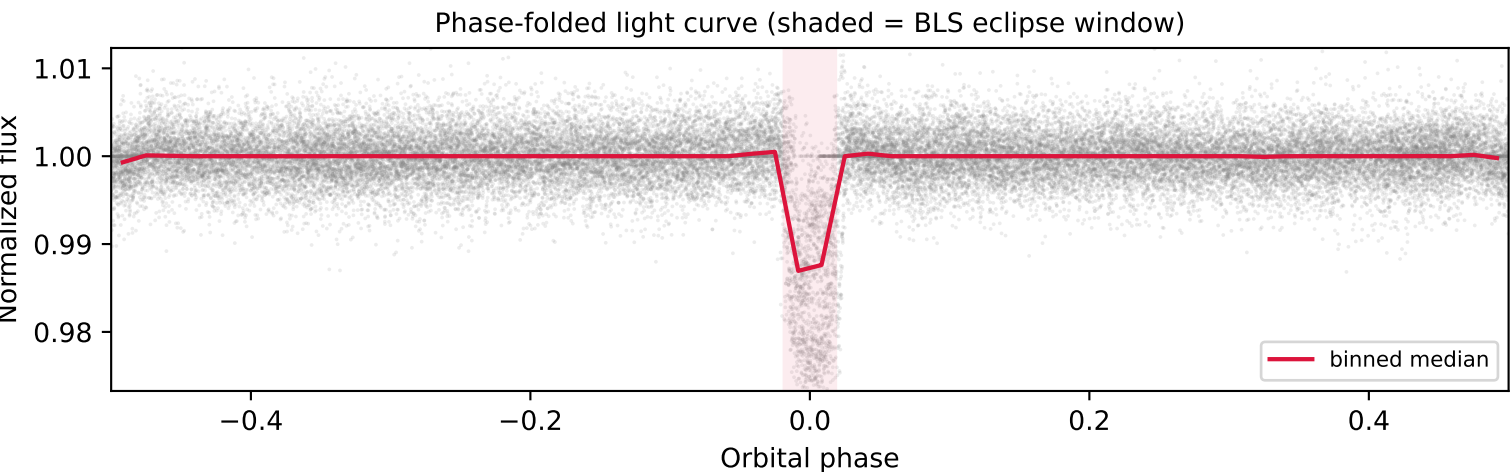
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 35 (gate >= 3)
Robust depth ratio 0.94 (gate >= 0.50)
Robust depth sigma 153.4 (gate >= 7.0)
Odd/even z-score 12.25 (ALIAS FLAGGED)
Secondary eclipse detected, phase -0.49, 5.7 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 74 light-curve dip in this sector: 61.0 sigma t0 refined by 0.0075 phase
Difference-image peak SNR 30.32 (gate >= 5.0) Compactness 0.85 (gate >= 0.50)
Dimming centroid offset 0.12 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 313485020
2.73 px away, dTmag +3.67 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 12.25 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 13.152313 d, in any write-up.